

Prop varianza sum var aleatorias indep

Proposición 1. Sean $(X_j)_{j=1}^n \subset \mathcal{L}^2(\mathbb{P})$ variables aleatorias independientes

$$\implies \mathbb{V}\left(\sum_{j=1}^n X_j\right) = \sum_{j=1}^n \mathbb{V}(X_j).$$

Demostración: Usamos la fórmula para la varianza:

$$\mathbb{V}\left(\sum_{j=1}^n X_j\right) = \mathbb{E}\left[\left(\sum_{j=1}^n X_j\right)^2\right] - \mathbb{E}\left[\sum_{j=1}^n X_j\right]^2 \stackrel{\uparrow \text{lineal}}{=} \mathbb{E}\left[\sum_{i,j=1}^n X_i \cdot X_j\right] - \left(\sum_{j=1}^n \mathbb{E}[X_j]\right)^2$$

Ahora bien, como $\forall i \neq j : X_i$ y X_j son independientes, por la **prop-esperanza-prod-var-aleatorias-indep** tenemos que $\mathbb{E}[X_i \cdot X_j] = \mathbb{E}[X_i] \cdot \mathbb{E}[X_j]$. Luego para el primer término tenemos:

$$\mathbb{E}\left[\sum_{i,j=1}^n X_i \cdot X_j\right] \stackrel{\downarrow \text{lineal}}{=} \sum_{j=1}^n \mathbb{E}[X_j^2] - \sum_{\substack{i,j=1 \\ i \neq j}}^n \mathbb{E}[X_i \cdot X_j] = \sum_{j=1}^n \mathbb{E}[X_j^2] - \sum_{\substack{i,j=1 \\ i \neq j}}^n \mathbb{E}[X_i] \cdot \mathbb{E}[X_j].$$

Por otro lado, para el segundo término, tenemos que

$$\left(\sum_{j=1}^n \mathbb{E}[X_j]\right)^2 = \sum_{i,j=1}^n \mathbb{E}[X_i] \cdot \mathbb{E}[X_j].$$

Por tanto, al restar el segundo término al primero, obtenemos:

$$\mathbb{V}\left(\sum_{j=1}^n X_j\right) = \sum_{j=1}^n \mathbb{E}[X_j^2] - \sum_{j=1}^n \mathbb{E}[X_j]^2 = \sum_{j=1}^n \mathbb{E}[X_j^2] - \mathbb{E}[X_j]^2.$$

Usando de nuevo la fórmula para la varianza, concluimos que $\mathbb{V}\left(\sum_j X_j\right) = \sum_j \mathbb{V}(X_j)$. ■

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